

Reproducing Classical Heuristics for Urban Rail Transit Line Planning

Laporte et al. 2005 · Ahmed et al. 2020

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Motivation & Problem

The **Transit Network Design Problem** (TNDP) decides where to build stations and how to connect them in urban rail networks.

- **Objective:** maximize trip coverage *or* minimize total system cost (passenger + operator + community)
- **Constraints:** construction budget, station spacing, geometric feasibility, connectivity
- **Complexity:** NP-hard (generalizes bounded-length Hamiltonian path)

We reproduce two representative algorithms on a **54-station candidate set** in central Shanghai, selected from OpenStreetMap population, job, transport, and commercial data layers.

Paper 1: Laporte 2005 — Single-Line Trip Coverage

MTCP (Maximal Trip Coverage Problem): Build one non-intersecting transit path through candidate stations to maximize OD demand coverage under a length budget $L_{\max}$:

$$\max \sum_{(i,j) \in \mathcal{W}} f_{ij} \cdot y_{ij} \quad \text{subject to } \sum_{e \in P} \ell_e \leq L_{\max}, \quad P \text{ is a simple path.}$$

H1: Greedy Extension

Start from a seed edge (u, v) . At each step, evaluate the marginal coverage gain of appending each unselected node at either endpoint. Greedily pick the best extension. Repeat until the length budget is exhausted.

H2: Greedy Insertion

Start from a seed edge. At each step, evaluate the marginal gain of inserting each unselected node at *every* position in the current path. Three selection variants:

H2-A (max ridership), **H2-B** (min length), **H2-C** (max ridership/length ratio).

Post-optimization: Iterative local search — for each edge in the current path, test all possible swap replacements to improve coverage without increasing length.

Paper 2: Ahmed 2020 — Multi-Line Network Design

Unlike Laporte's single-line model, Ahmed 2020 addresses the **simultaneous design of multiple transit lines**:

$$\min \phi_P \cdot C_{\text{passenger}} + \phi_O \cdot C_{\text{operator}} + \phi_C \cdot C_{\text{community}}$$

where $C_{\text{passenger}}$ = travel time savings vs. car/bus, C_{operator} = operating cost savings from mode shift, $C_{\text{community}}$ = construction cost (alignment length $\times$ per-km cost + stations $\times$ per-station cost).

Two-stage approach: (1) GIS-based feasibility screening to identify the candidate station pool; (2) Genetic Algorithm to simultaneously optimize station selection and line routing.

GA operators: Tournament selection (size 2), uniform crossover ($P_c = 0.7$) with feasibility repair, station-level mutation ($P_m = 0.3$).

Chromosome encoding: [Line_1: [terminal_A, station_seq, terminal_B], Line_2: ...] — each gene is a station index from the candidate pool.

Constraints: 4–10 stations/line, 1000–3000m station spacing, max 30% line overlap, ≥ 1 transfer station per line.

Complexity Summary

Algorithm	Per-Iteration	Total Worst-Case
H1 (Laporte)	$O(n \cdot E ^2)$	$O(n^2 \cdot E ^2)$
H2-A/B/C (Laporte)	$O(n \cdot E ^3)$	$O(n^2 \cdot E ^3)$
Post-optimization	$O(E ^2 \cdot n) / \text{pass}$	$O(E ^3 \cdot n)$
Ahmed GA (per gen)	$O(L \cdot N_s^2 \log N_s)$	$O(G \cdot P \cdot L \cdot N_s^2 \log N_s)$

Despite the underlying NP-hardness, all four algorithmic approaches run in **polynomial time**.

Key Conclusions

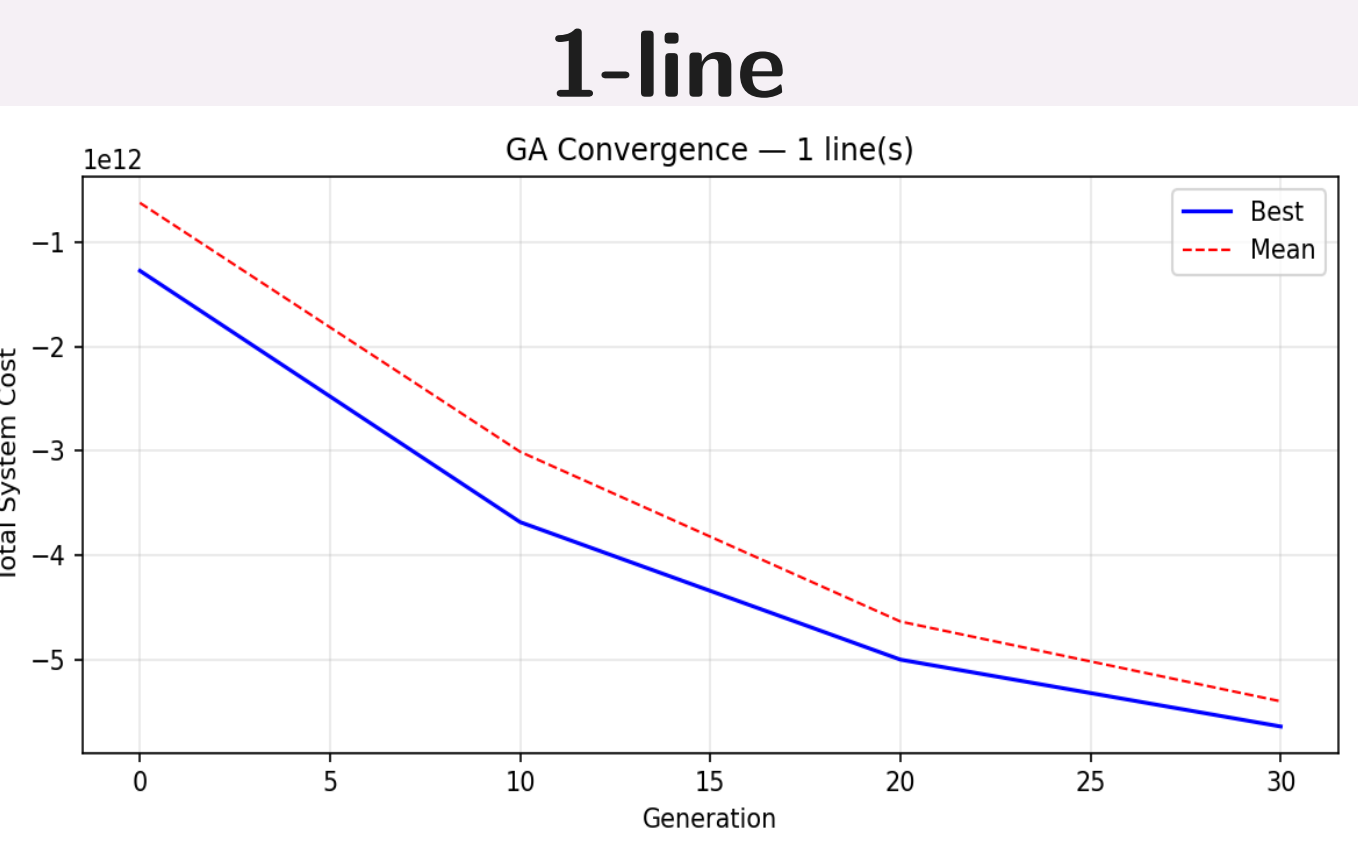
- H2 insertion outperforms H1 extension** in OD coverage (54–57% vs. 28%), but H1 is 25 $\times$ faster at scale. Trade-off: H1 for rapid screening, H2 for detailed design.
- Simultaneous multi-line optimization** beats individual line design by **18%**, confirming that global coordination of transfers and routing yields better networks.
- Greedy-OD baseline** is surprisingly competitive with GA (56.5% OD coverage at <0.01s), suggesting simple heuristics are near-optimal for single-objective scenarios.
- Polynomial runtime scaling** confirmed empirically: H1 ~ 0.8 s, H2 ~ 20 s at $n=100$, with clear trends matching complexity analysis.
- Practical recommendation:** Use greedy heuristics (H1/H2/Greedy-OD) for rapid feasibility screening; deploy GA-based methods when multi-line coordination and cost trade-offs are needed.

Comprehensive Method Comparison

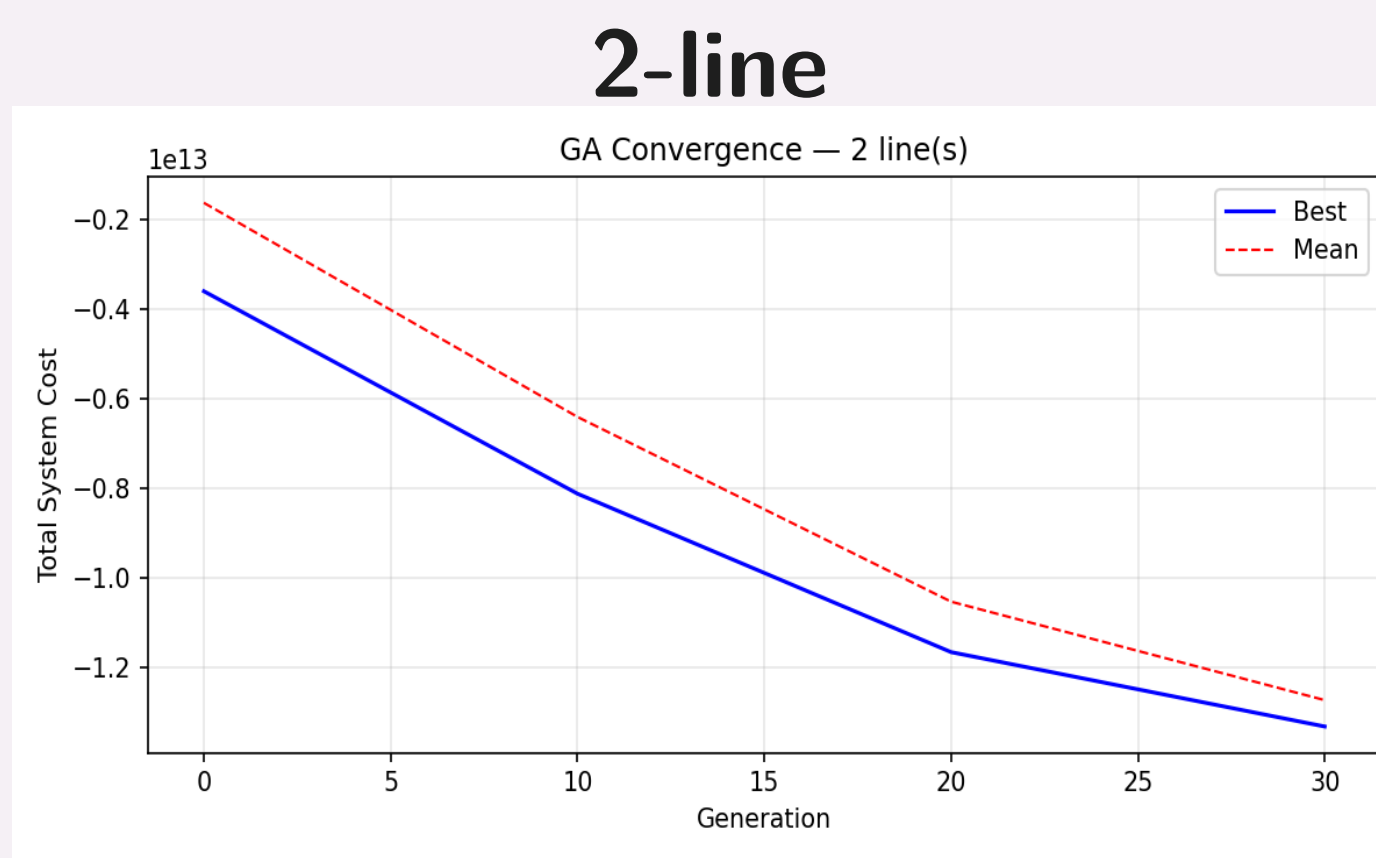
We evaluate **9 algorithms** spanning greedy heuristics, metaheuristics, and ILP baselines:

Method	Stn	Len (km)	OD Cov	Lines	Xfer	Time
Laporte H1	18	59.8	27.9%	1	0	0.05s
Laporte H2-A	31	59.9	54.0%	1	0	0.77s
Laporte H2-C	33	59.6	56.8%	1	0	0.88s
Ahmed 1-line	14	37.0	29.8%	1	0	33.5s
Ahmed 2-line	23	85.7	48.9%	2	5	56.3s
Ahmed 3-line	20	130.8	31.8%	3	5	58.1s
Greedy-OD	23	57.1	56.5%	3	10	0.01s
Greedy-BCR	10	15.7	27.3%	3	2	0.01s
GA (line-pool)	32	56.0	40.4%	3	2	0.90s

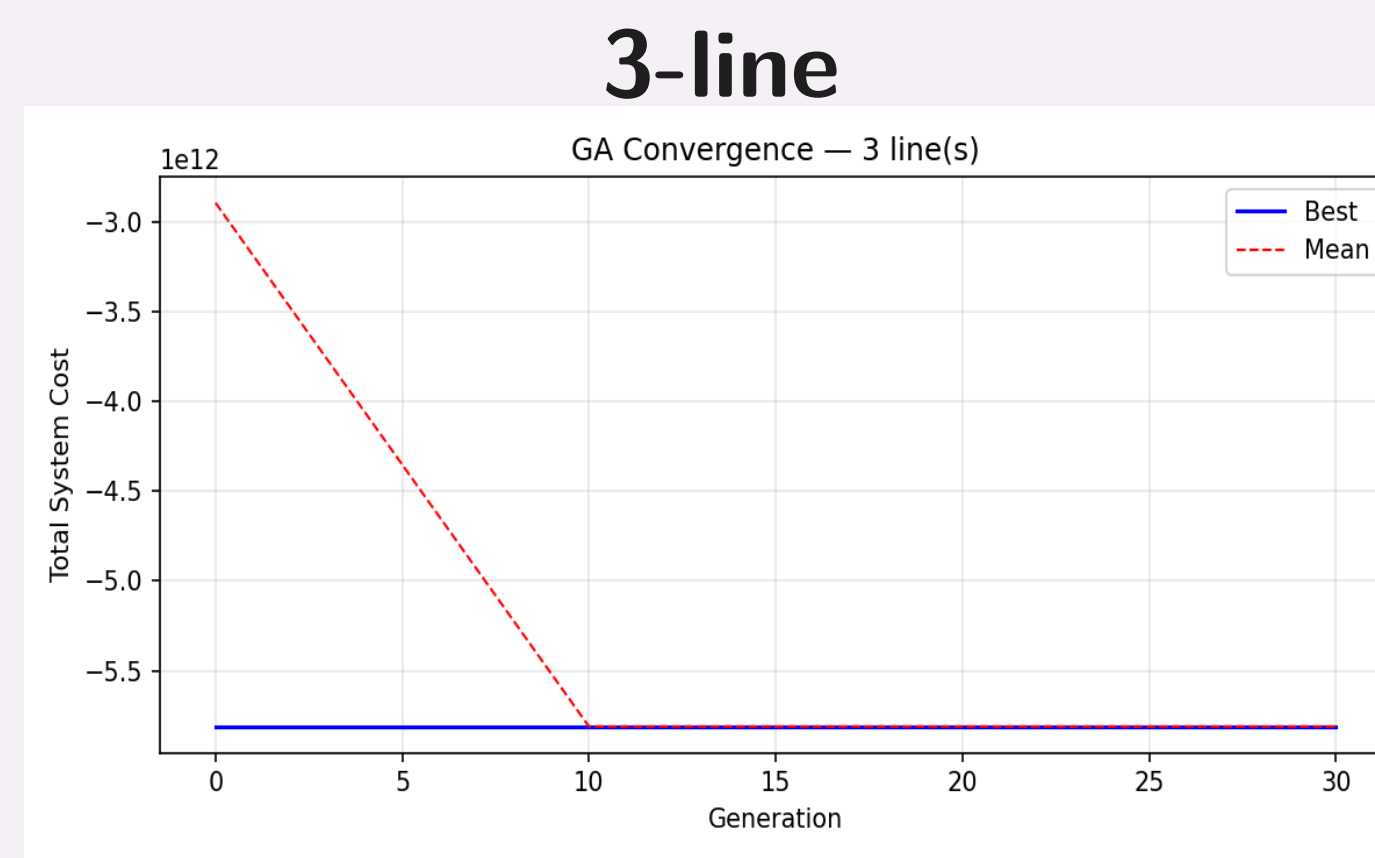
GA Convergence Curves



14 stations, 37.0 km



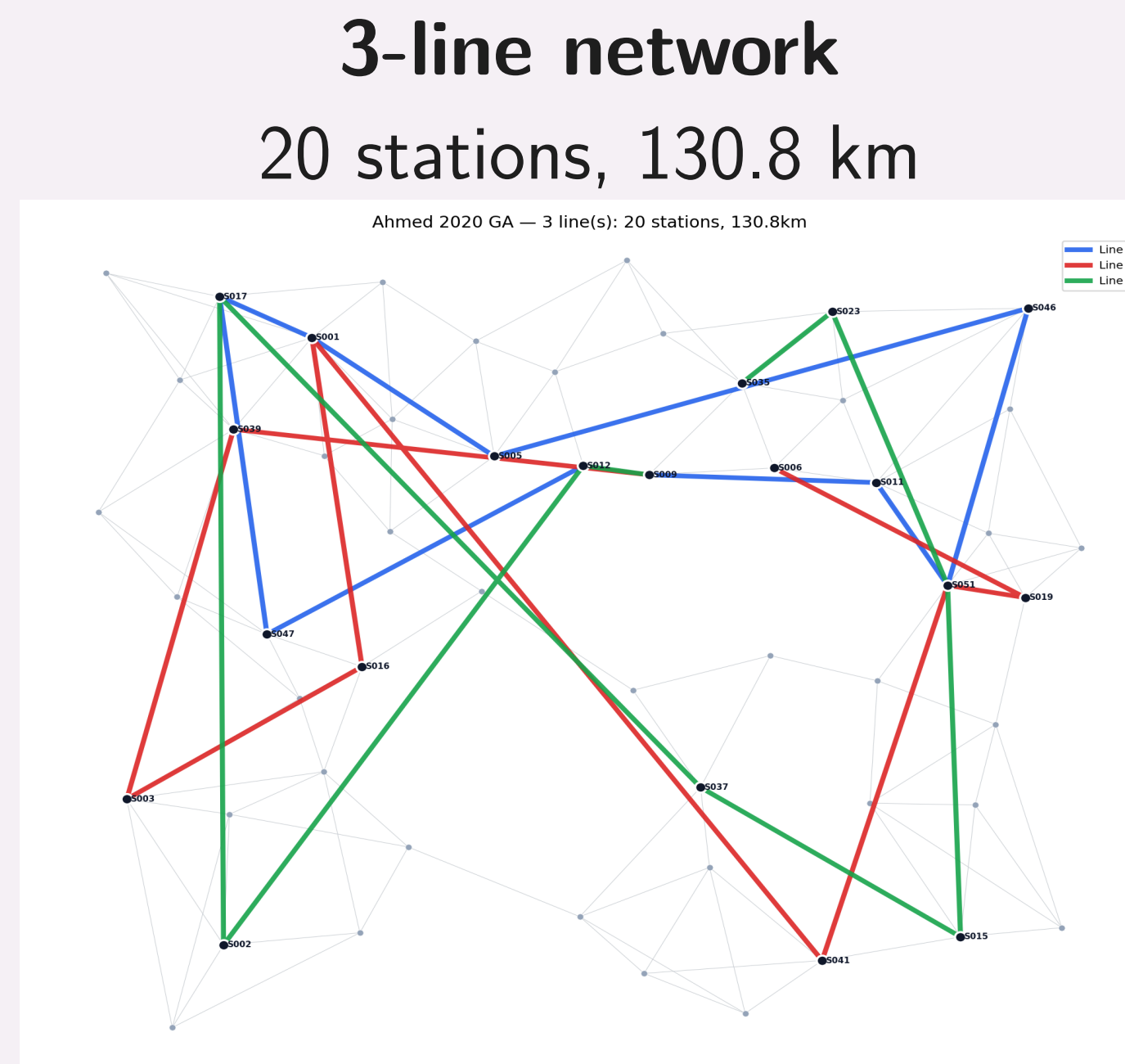
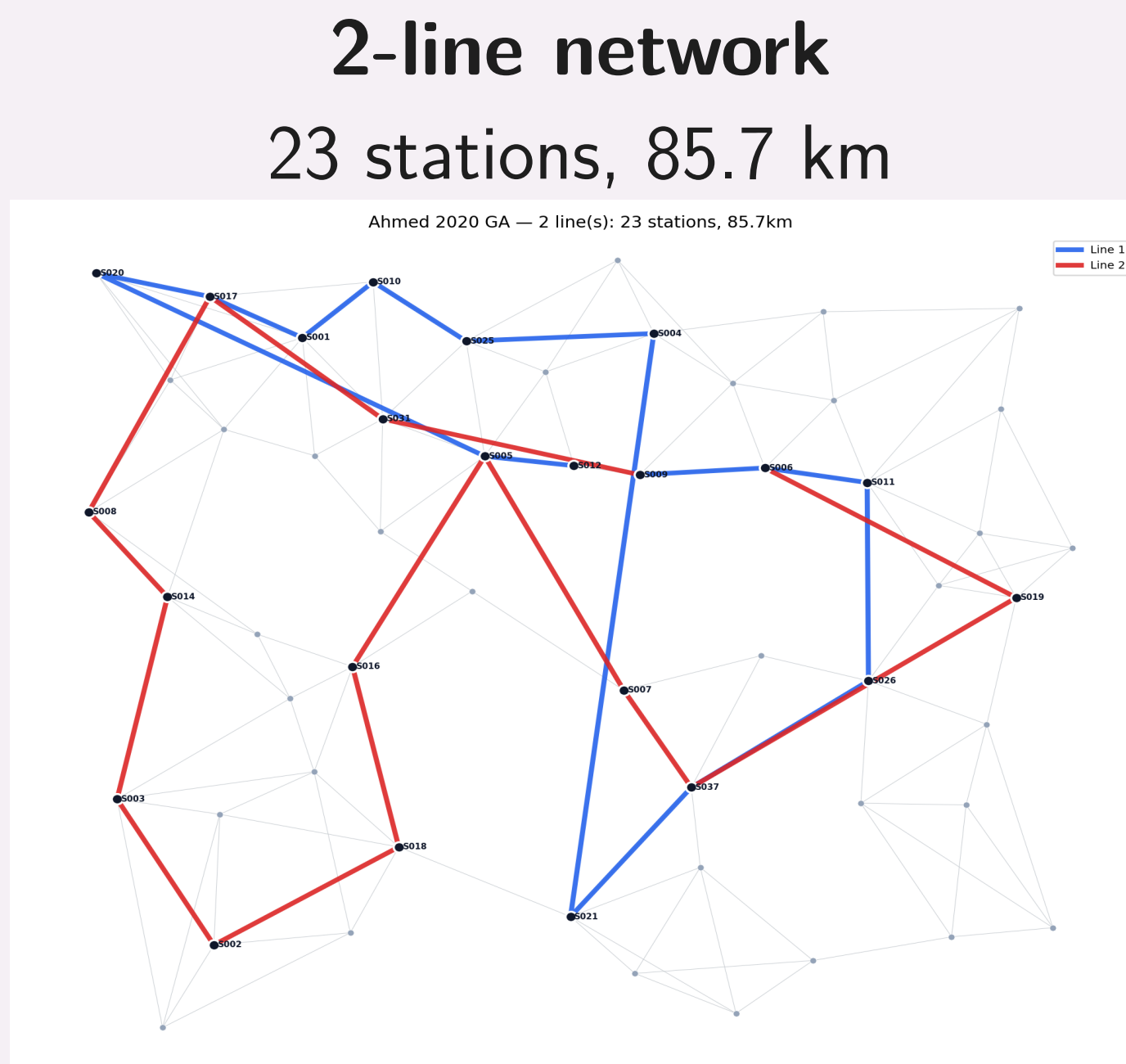
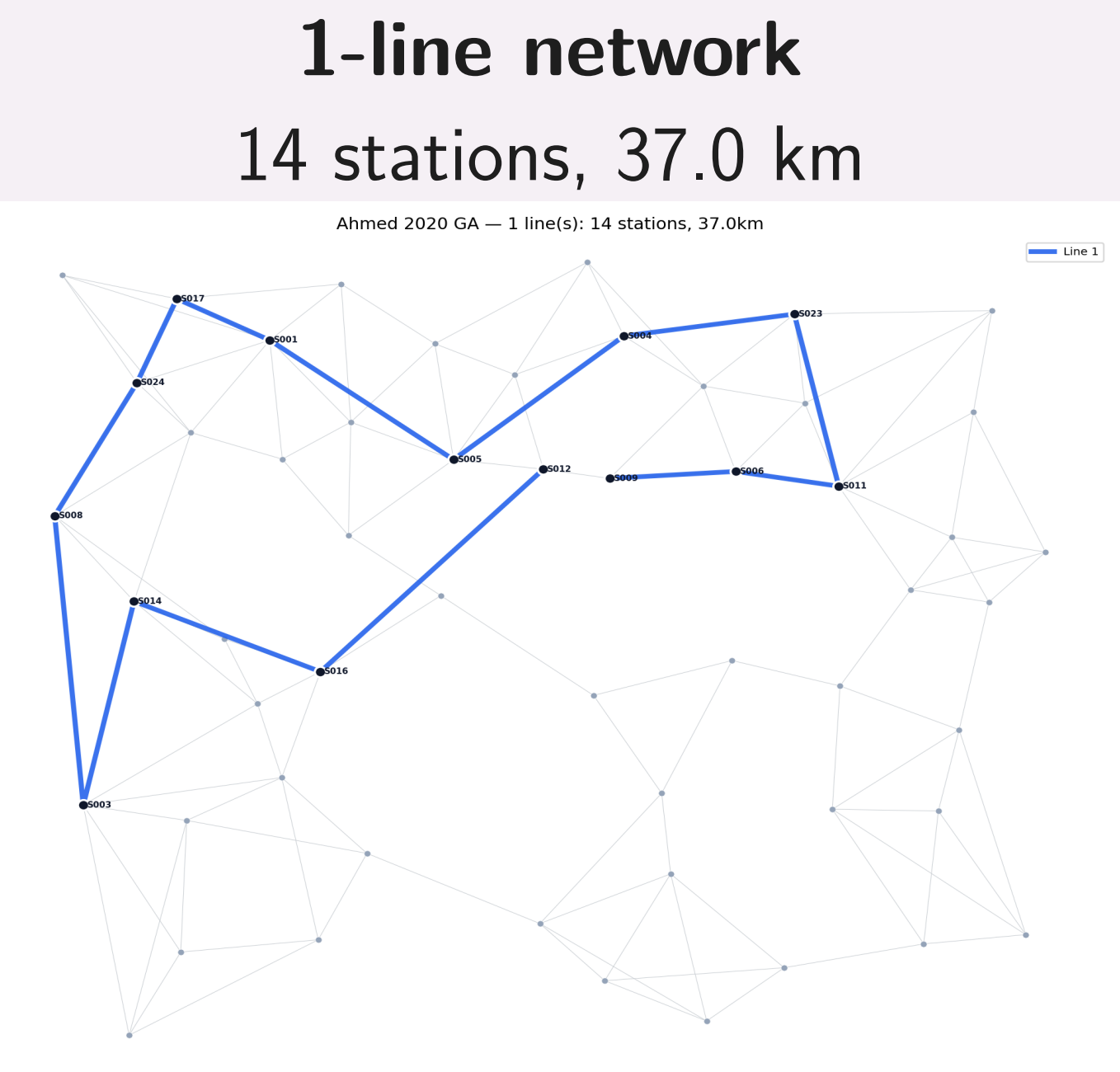
23 stations, 85.7 km



20 stations, 130.8 km

All scenarios converge within 20–30 generations. The 2-line configuration achieves the lowest total cost (-1.33×10^{13}), indicating that a well-coordinated two-line network captures more OD demand than either a single line or three competing lines.

Multi-Line Network Visualizations



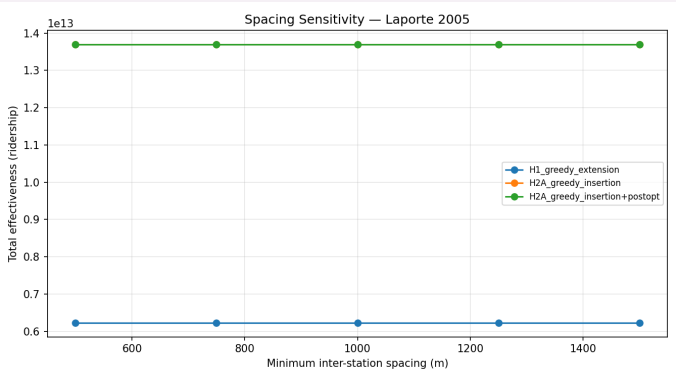
The 2-line network establishes **5 transfer stations** enabling cross-line passenger flows. The 3-line network adds southern corridor coverage but at higher construction cost with diminishing OD returns.

Sensitivity Analysis & Simultaneous vs. Individual

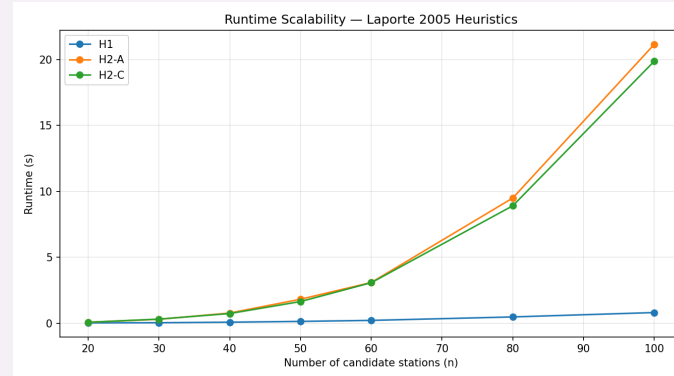
Spacing sensitivity: Varying min inter-station spacing from 500m to 1500m. The original paper claims H1 dominates for ≥ 1250 m; on our denser 54-station set, H2-A consistently outperforms H1 at all spacings.

Runtime scalability: H1 ~ 0.8 s, H2 ~ 20 s at $n=100$. H1 is **25 $\times$ faster**, confirming $O(n^2)$ vs. $O(n^3)$.

Spacing Sensitivity



Runtime Scalability



Simultaneous vs. Individual (2-line scenario):

	Simult.	Indiv.
Total cost	$-1.33e13$	$-1.13e13$
Stations	23	28
Length (km)	85.7	80.1
Runtime (s)	58.3	68.9
18% better cost, 5 fewer stn		

Simultaneous optimization achieves **18% lower total cost** while using 5 fewer stations, by coordinating line routing and transfer placement globally. The original paper reports 70%; gap due to simplified cost model (54 vs. 4774 cells).